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(57) Abstract :
ABSTRACT A COMPUTER-IMPLEMENTED SYSTEM FOR AUTOMATED BRAIN TUMOR DETECTION The invention provides a computer-implemented system (100) for automated brain tumor detection and explainable classification from MRI images. The system includes an input unit (101) for acquiring MRI data, a preprocessing module (102) for performing resizing, normalization, denoising, or augmentation, and a classification module (103) employing a modified ResNet-50 deep learning architecture. An optimization module (104) utilizes a Grey Wolf Optimization (GWO) algorithm to adaptively tune hyperparameters for improved accuracy and convergence. An explainability module (105) generates Grad-CAM heatmaps to highlight tumor-relevant regions, and an output module (106) provides the predicted tumor class along with the corresponding visualization. The system delivers enhanced diagnostic performance, improved computational efficiency, and clinically interpretable results for multi-class brain tumor identification. [to be published with figure 1]

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